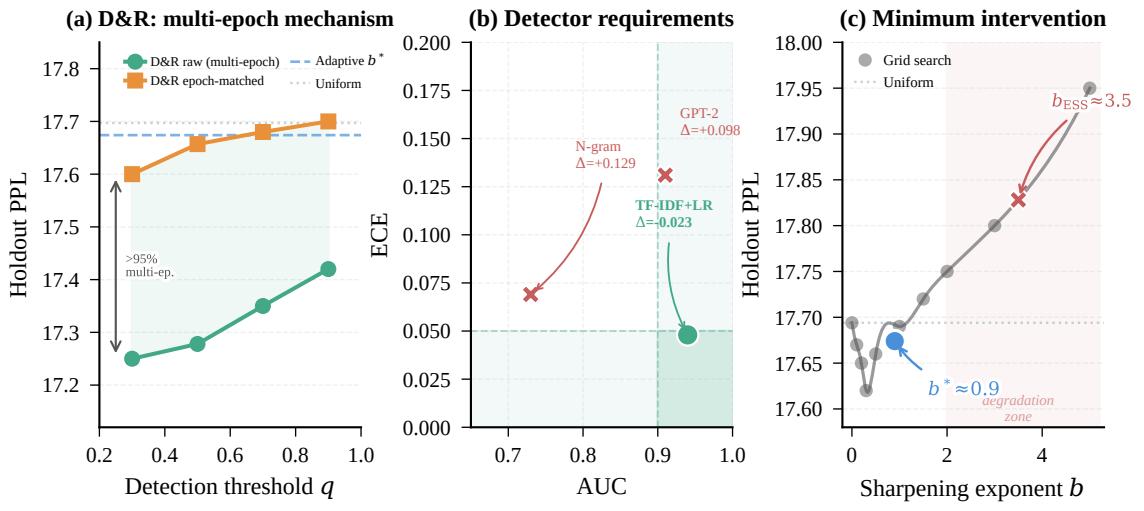
(a) D&R: multi-epoch mechanism
D&R raw (multi-epoch)
D&R epoch-matched
Adaptive b^*
Uniform
>95% multi-ep.
Holdout PPL
Detection threshold q
(b) Detector requirements
N-gram
Δ=+0.129
GPT-2
Δ=+0.098
TF-IDF+LR
Δ=-0.023
ECE
AUC
(c) Minimum intervention
Grid search
Uniform
$b_{\mathrm{ESS}} \approx 3.5$
$b^* \approx 0.9$
degradation zone
Holdout PPL
Sharpening exponent b